THE HYPERCOVERING CECH COMPLEX AS HOM FROM THE NORMALIZED COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs lemma-push-pull-localization, lemma-augmentation-cech-complex; the normalized complex s(Z_{F(K)}^#) from Hypercoverings.

Editorial note. The label lemma-hypercovering-cech-complex names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Push-pull on localization. Same unit, Lemma lemma-push-pull-localization. Term identification: a_{n,*}a_n^{-1}F = SheafHom'(F(K_n)^#, F).
- Augmentation Cech complex. Same unit, Lemma lemma-augmentation-cech-complex. The complex whose terms are identified; its boundary maps come from the simplicial structure.
- Normalized complex. Hypercoverings, Definition hypercovering-definition-homology. Defines s(Z_{F(K)}^#), the target of the identification.

Source reference. Original definition 01GB: sources/stacks-chow/tags.txt, source lines 1900–1900. Prerequisite pointer invoked at proof line(s) 4365; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D79: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1988–1988. Prerequisite pointer invoked at proof line(s) 4357; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D86: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3708–3708. Prerequisite pointer invoked at proof line(s) 4369; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 04TP: sources/stacks-chow/tags.txt, source lines 6262–6262. Prerequisite pointer invoked at proof line(s) 4373; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a site with equalizers and fibre products. Let K be a hypercovering. The Čech complex of Lemma 0D79 associated to $a^{-1}\mathcal{F}$*

$$a_{0,*}a_0^{-1}\mathcal{F} \to a_{1,*}a_1^{-1}\mathcal{F} \to a_{2,*}a_2^{-1}\mathcal{F} \to \ldots$$

is equal to the complex $\mathcal{H}om(s(\mathbf{Z}^{\#}_{F(K)}), \mathcal{F})$. Here $s(\mathbf{Z}^{\#}_{F(K)})$ is as in Hypercoverings, Definition 01GB.

Proof. By Lemma 0D86 we have

$$a_{n,*}a_n^{-1}\mathcal{F} = \mathcal{H}om'(F(K_n)^{\#}, \mathcal{F})$$

where $\mathcal{H}om'$ is as in Sites, Section 04TP. The boundary maps in the complex of Lemma 0D79 come from the simplicial structure. Thus the equality of complexes comes from the canonical identifications $\mathcal{H}om'(\mathcal{G}, \mathcal{F}) = \mathcal{H}om(\mathbf{Z}_{\mathcal{G}}, \mathcal{F})$ for $\mathcal{G}$ in $Sh(\mathcal{C})$. □